Strongly Connected Edges

Time limit: 1.00 s **Memory limit:** 512 MB

Given an undirected graph, your task is to choose a direction for each edge so that the resulting directed graph is strongly connected.

Input

The first input line has two integers n and m: the number of nodes and edges. The nodes are numbered $1, 2, \ldots, n$.

After this, there are m lines describing the edges. Each line has two integers a and b: there is an edge between nodes a and b.

You may assume that the graph is simple, i.e., there are at most one edge between two nodes and every edge connects two distinct nodes.

Output

Print m lines describing the directions of the edges. Each line has two integers a and b: there is an edge from node a to node b. You can print any valid solution.

If there are no solutions, only print `IMPOSSIBLE`.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
3 3 1 2 1 3 2 3	1 2 2 3 3 1